Complete Error Reference Manual

Student Information System (SIS)

CSE 102 Programming Project

January 2026

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1 Input Validation Errors

1.1 Invalid Numeric Input

Error: `"Invalid input."`
Cause: User entered non-numeric characters when a number was expected.
Solution: Enter only digits (0-9). Do not include letters, spaces, or special characters.
Example: When prompted "Enter student ID:", type `2024001`, not `ABC123` or `two thousand`.

1.2 Invalid Student ID Format

Error: `"Invalid Student ID format. ID must be 7 digits (e.g., 2024001)."`
Cause: Student ID does not match the required 7-digit YYYYXXX format.
Solution: Enter exactly 7 digits where first 4 digits represent year (2000-2026) and last 3 digits are sequence number (001-999).
Example: Type `2024001`, not `202401` (6 digits) or `20240001` (8 digits).

1.3 Invalid Student ID Year Range

Error: `"Invalid Student ID format. Year part (first 4 digits) must be 2000-2026."`
Cause: The year portion of Student ID is outside the valid range.
Solution: Use a year between 2000 and 2026 for the first 4 digits.
Example: Type `2024001`, not `1999001` (too old) or `2030001` (too future).

1.4 Invalid Student ID Sequence Number

Error: `"Invalid Student ID format. Sequence number (last 3 digits) cannot be 000."`
Cause: The sequence number (last 3 digits) is 000, which is not allowed.
Solution: Use sequence numbers from 001 to 999.
Example: Type `2024001`, not `2024000`.

1.5 ID Cannot Be Zero or Negative

Error: `"ID cannot be zero or negative."`
Cause: User entered 0 or a negative number for an ID field.
Solution: Enter a positive integer greater than zero.
Example: Type `2024001`, not `0` or `-1`.

1.6 Invalid Name Format

Error: `"Name must contain only alphabetic characters!"`
Cause: Name field contains numbers or special characters.
Solution: Use only letters (A-Z, a-z), spaces, and hyphens. No numbers or symbols allowed.
Example: Type `John`, not `John123` or `John@Smith`.

1.7 Invalid Email Format - Missing Domain

Error: `"Email must end with @university.edu"`
Cause: Email address does not have the required domain suffix.
Solution: Use email format: `name.surname@university.edu`
Example: Type `john.doe@university.edu`, not `john.doe@gmail.com`.

1.8 Invalid Email Format - Wrong Structure

Error: `"Email format must be name.surname (only one dot allowed)."`
Cause: Email has incorrect format before the @ symbol.
Solution: Use exactly one dot between name and surname.
Example: Type `john.doe@university.edu`, not `johndoe@university.edu`.

1.9 Invalid Email Format - Invalid Characters

Error: `"Email contains invalid characters. Use lowercase only."`
Cause: Email contains uppercase letters or special characters.
Solution: Use only lowercase letters, one dot, and the required domain.
Example: Type `john.doe@university.edu`, not `John.Doe@university.edu`.

1.10 Invalid Phone Number Format

Error: `"Phone number must be exactly 8 characters (555-XXXX)."`
Cause: Phone number does not match required format.
Solution: Use format `555-XXXX` where X is a digit.
Example: Type `555-1234`, not `5551234`.

1.11 Invalid Phone Number Prefix

Error: `"Phone number must start with 555."`
Cause: Phone number does not start with the required 555 prefix.
Solution: All phone numbers must begin with 555.
Example: Type `555-1234`, not `123-4567`.

1.12 Invalid Phone Number Hyphen

Error: `"Phone number must contain a hyphen at 4th position."`
Cause: Phone number is missing the hyphen or has it in wrong position.
Solution: Place hyphen after the first 3 digits: `555-XXXX`.
Example: Type `555-1234`, not `5551234`.

1.13 Invalid GPA Value

Error: `"GPA must be between 0.0 and 4.0."`
Cause: GPA value is outside the valid range.

Solution: Enter a decimal number between 0.0 and 4.0.
Example: Type `3.5`, not `4.5` or `-1.0`.

1.14 Invalid Semester Format - General

Error: `"Invalid format. Example: 2021-SPRING (Year-Season)."`
Cause: Semester does not match YYYY-SEASON format.
Solution: Use format: `YYYY-SEASON` with hyphen separator.
Example: Type `2024-FALL`, not `Fall 2024`.

1.15 Invalid Semester Year

Error: `"Semester year must start with 4 digits (e.g. 2021)."`
Cause: Year portion of semester is not a valid 4-digit number.
Solution: Use a 4-digit year like 2024.
Example: Type `2024-FALL`, not `24-FALL`.

1.16 Invalid Semester Hyphen

Error: `"Hyphen (-) must follow the year. E.g., 2021-SPRING"`
Cause: Missing or misplaced hyphen in semester format.
Solution: Place hyphen immediately after the 4-digit year.
Example: Type `2024-FALL`, not `2024FALL`.

1.17 Invalid Semester Season

Error: `"Season must be SPRING, FALL, or SUMMER (case-insensitive)."`
Cause: Invalid season name used.
Solution: Use only SPRING, FALL, or SUMMER after the hyphen.
Example: Type `2024-FALL`, not `2024-WINTER`.

1.18 Invalid Date Format

Error: `"Date must be in YYYY-MM-DD format (10 characters)."`
Cause: Date does not match the required ISO format.
Solution: Use format: `YYYY-MM-DD` (e.g., 2024-01-15).
Example: Type `2024-01-15`, not `01/15/2024`.

1.19 Invalid Day for Month

Error: `"Invalid day for the given month."`
Cause: Day number exceeds the maximum for that month.
Solution: Check the correct number of days for the month (28/29/30/31).
Example: Type `2024-02-29` (leap year), not `2024-02-30`.

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2 Student Module Errors

2.1 Student Not Found

Error: `"Student not found with this ID."`
Cause: The entered Student ID does not exist in the database.
Solution: Verify the Student ID using Search function, or add the student first.
Example: Before editing 2024999, search to confirm it exists.

2.2 Duplicate Student ID

Error: `"Student with this ID already exists. Please try another."`
Cause: Attempting to add a student with an ID that's already registered.
Solution: Use a different Student ID or edit the existing student record.
Example: If 2024001 exists, try 2024002.

2.3 Invalid Student ID Input

Error: `"Invalid input. Student ID must be numeric."`
Cause: Non-numeric characters entered for Student ID.
Solution: Enter only digits for Student ID.
Example: Type `2024001`, not `STU-2024-001`.

2.4 Student File Parse Error - Column Count

Error: `"[ERROR] Line X: Invalid column count. Skipped."`
Cause: CSV row has wrong number of columns (expected 8).
Solution: Check students.csv file and ensure each row has exactly 8 comma-separated values.
Example: Correct: `2024001,John,Doe,...`

2.5 Student File Parse Error - Invalid Name

Error: `"[ERROR] Line X: Invalid First Name 'John123'. (Contains numbers/symbols). Skipped."`
Cause: Name in CSV file contains invalid characters.
Solution: Edit the CSV file and remove numbers/symbols from name fields.
Example: Change `John123` to `John`.

2.6 Student File Parse Error - Invalid Major

Error: `"[ERROR] Line X: Invalid Major Name 'CS101'. Skipped."`
Cause: Major field in CSV contains numbers or invalid characters.
Solution: Use text-only major names.
Example: Change `CS101` to `Computer Science`.

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3 Professor Module Errors

3.1 Professor Not Found

Error: `"Professor not found with this ID."`
Cause: The entered Professor ID does not exist in the database.
Solution: Verify the Professor ID using List function.
Example: Before assigning professor 5001, confirm existence.

3.2 Duplicate Professor ID

Error: `"Professor with this ID already exists!"`
Cause: Attempting to add a professor with an ID that's already registered.
Solution: Use a different Professor ID.
Example: If 5001 exists, try 5002.

3.3 Invalid Professor ID Range

Error: `"Invalid Professor ID. ID must be greater than 1000."`
Cause: Professor ID is too small.
Solution: Use Professor IDs between 1001 and 9999.
Example: Type `5001`, not `999`.

3.4 Invalid Professor ID Maximum

Error: `"Invalid Professor ID. ID must be less than 9999."`
Cause: Professor ID exceeds maximum value.
Solution: Use Professor IDs less than 9999.
Example: Type `5001`, not `10000`.

3.5 Professor ID Cannot Be Negative

Error: `"Professor ID cannot be zero or negative."`
Cause: User entered 0 or negative number.
Solution: Enter a positive integer.
Example: Type `5001`, not `-5001`.

3.6 Search Term Empty

Error: `"Search term cannot be empty."`
Cause: User pressed Enter without typing a search term.
Solution: Enter at least one character for the search.
Example: Type `Smith`, not just pressing Enter.

3.7 Invalid Menu Option

Error: `"Invalid menu option. Please enter 0-6."`
Cause: User entered a number outside the valid menu range.
Solution: Enter a number corresponding to a menu option.
Example: Type `1` for Add, not `9`.

3.8 Professor File Parse Error

Error: `"[ERROR] File: X, Line: Y -> Invalid format. Expected 8 columns, found Z. Skipped."`
Cause: CSV row has incorrect number of columns.
Solution: Ensure each row in professors.csv has exactly 8 columns.
Example: Correct: `5001,John,Smith,...`

3.9 Invalid Professor Email in CSV

Error: `"[ERROR] Line X: Invalid Email Format (abc@gmail.com). Skipped."`
Cause: Email in CSV doesn't match required university format.
Solution: All professor emails must end with @university.edu.
Example: Change `abc@gmail.com` to `john.smith@university.edu`.

3.10 Invalid Professor Phone in CSV

Error: `"[ERROR] Line X: Invalid Phone Format (123-4567). Skipped."`
Cause: Phone number in CSV doesn't match 555-XXXX format.
Solution: All phone numbers must start with 555.
Example: Change `123-4567` to `555-4567`.

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4 Course Module Errors

4.1 Course Not Found

Error: `"Course not found with this ID."`
Cause: The entered Course ID does not exist in the database.
Solution: Verify the Course ID using List function.
Example: Check ID 1001 before enrolling.

4.2 Course Not Found (Alternate)

Error: `"Course not found."`
Cause: Course ID provided to a function does not match any course.
Solution: Use Course Management Search to find the correct Course ID.
Example: Search for "CSE101" to find its ID.

4.3 Duplicate Course ID

Error: `"Course ID already exists."`
Cause: Attempting to add a course with an ID that's already in use.
Solution: Use a different Course ID.
Example: If 1001 exists, try 1002.

4.4 Duplicate Course Code

Error: `"Course Code already exists! Please use a unique code."`
Cause: Attempting to add a course with a code that's already registered.
Solution: Use a unique course code.
Example: If "CSE101" exists, try "CSE102".

4.5 Invalid Course ID Format

Error: `"Invalid Course ID! ID must be exactly 4 digits (1000-9999)."`
Cause: Course ID is not a 4-digit number.
Solution: Use Course IDs between 1000 and 9999.
Example: Type `1001`, not `101`.

4.6 Invalid Course ID Input

Error: `"Invalid input. ID must be strictly numeric."`
Cause: Non-numeric characters entered for Course ID.
Solution: Enter only digits.
Example: Type `1001`, not `CSE101`.

4.7 Invalid Credits Format

Error: `"Invalid input. Credits must be a number."`
Cause: Non-numeric input for credits field.
Solution: Enter a whole number for credits.
Example: Type `3`, not `three`.

4.8 Negative Credits

Error: `"Credits cannot be negative."`
Cause: Negative number entered for credits.
Solution: Enter a positive number (typically 1-6).
Example: Type `3`, not `-3`.

4.9 Invalid Capacity Format

Error: `"Invalid input. Capacity must be a number."`
Cause: Non-numeric input for capacity field.

Solution: Enter a whole number for capacity.
Example: Type `30`, not `thirty`.

4.10 Negative Capacity

Error: `"Capacity cannot be negative."`
Cause: Negative number entered for capacity.
Solution: Enter a positive number.
Example: Type `30`, not `-30`.

4.11 Self-Referencing Prerequisite

Error: `"A course cannot be its own prerequisite."`
Cause: Attempting to add a course as its own prerequisite.
Solution: Select a different course as prerequisite.
Example: CSE101 cannot require CSE101.

4.12 Circular Prerequisite

Error: `"Circular logic: A course cannot be its own prerequisite."`
Cause: Adding prerequisite would create a circular dependency.
Solution: Review the prerequisite chain and avoid loops.
Example: If A requires B, B cannot require A.

4.13 Invalid Professor Assignment

Error: `"Invalid Professor ID."`
Cause: Professor ID entered for course assignment doesn't exist.
Solution: Verify professor exists before assigning to course.
Example: Add professor 5001 before assigning.

4.14 Professor Department Mismatch

Error: `"This professor is not in the required department!"`
Cause: Assigned professor's department doesn't match course department.
Solution: Assign a professor from the same department as the course.
Example: CS courses need CS professors.

4.15 Invalid Choice in Edit Menu

Error: `"Invalid choice."`
Cause: User entered a number outside valid menu options during course edit.
Solution: Enter a valid menu option number.
Example: Choose 1-7 for fields or 0 to save.

4.16 Course File Parse Error - Non-numeric ID

Error: `"[ERROR] Line X: Course ID 'ABC' is not numeric. Skipped."`
Cause: Course ID in CSV is not a valid number.
Solution: Ensure Course ID column contains only digits.
Example: Change `ABC` to `1001`.

4.17 Course File Parse Error - Non-numeric Credits

Error: `"[ERROR] Line X: Credits 'three' contains non-numeric characters. Skipped."`
Cause: Credits field in CSV is not numeric.
Solution: Use numeric values for credits.
Example: Change `three` to `3`.

4.18 Course File Parse Error - Duplicate ID

Error: `"[ERROR] Line X: Duplicate Course ID (1001). Already loaded. Skipped."`
Cause: Same Course ID appears multiple times in CSV.
Solution: Remove duplicate entries from CSV file.
Example: Keep only one row with Course ID 1001.

4.19 Course File Parse Error - Duplicate Code

Error: `"[ERROR] Line X: Duplicate Course Code 'CSE101'. Already loaded. Skipped."`
Cause: Same course code appears multiple times in CSV.
Solution: Use unique course codes for each course.
Example: Rename duplicate to CSE102.

---

5 Enrollment Module Errors

5.1 Student Not Found for Enrollment

Error: `"Student not found."`
Cause: Student ID entered for enrollment doesn't exist.
Solution: Add the student first, then create enrollment.
Example: Add 2024999 before enrolling.

5.2 Course Not Found for Enrollment

Error: `"Course not found."`
Cause: Course ID entered for enrollment doesn't exist.
Solution: Add the course first, then create enrollment.
Example: Add course 9999 before enrolling.

5.3 Duplicate Enrollment

Error: `"Student is already enrolled in this course for this semester."`
Cause: Student already has an active enrollment in this course for the same semester.
Solution: Use a different semester or update the existing enrollment.
Example: If enrolled in 2024-FALL, use 2025-SPRING.

5.4 Course Full

Error: `"Course is full for this semester!"`
Cause: Course has reached its maximum capacity for the semester.
Solution: Choose a different section or wait for next semester.
Example: If CSE101 is 30/30, try 2025-SPRING.

5.5 Cannot Drop Completed Course

Error: `"Cannot drop a completed course!"`
Cause: Attempting to drop an enrollment that has "Completed" status.
Solution: Completed courses cannot be dropped by design.
Example: Contact admin for transcript modification.

5.6 Enrollment Record Not Found

Error: `"Enrollment record not found."`
Cause: Specified enrollment ID doesn't exist.
Solution: Use View Student Enrollments to find the correct enrollment ID.
Example: Verify ID 999 exists before updating.

5.7 Professor ID Not Found for Enrollment

Error: `"Professor ID not found."`
Cause: Professor ID specified for enrollment doesn't exist.
Solution: Add the professor first or use an existing professor ID.
Example: Verify professor 9999 exists.

5.8 Enrollment File Parse Error - Column Count

Error: `"[ERROR] Line X: Invalid column count. Skipped."`
Cause: Enrollment CSV row has wrong number of columns (expected 7).
Solution: Ensure each row has: enrollment_id, student_id, course_id, etc.
Example: Correct: `1,2024001,1001,...`

5.9 Enrollment File Parse Error - Non-numeric IDs

Error: `"[ERROR] Line X: Enrollment ID 'ABC' is not numeric. Skipped."`
Cause: ID fields in CSV contain non-numeric values.

Solution: Ensure all ID columns contain only numbers.
Example: Change `ABC` to `1`.

5.10 Enrollment File Parse Error - Duplicate ID

Error: `"[ERROR] Line X: Duplicate Enrollment ID (1). Skipped."`
Cause: Same enrollment ID appears multiple times.
Solution: Each enrollment must have a unique ID.
Example: Use unique IDs.

5.11 Enrollment File Parse Error - Invalid Status

Error: `"[ERROR] Line X: Invalid Status 'Active'. Must be 'Enrolled' or 'Completed'. Skipped."`
Cause: Status field has invalid value.
Solution: Use only "Enrolled" or "Completed".
Example: Change `Active` to `Enrolled`.

5.12 Enrollment File Parse Error - Invalid Semester

Error: `"[ERROR] Line X: Invalid Semester Format 'Fall 2024'. Skipped."`
Cause: Semester format doesn't match YYYY-SEASON.
Solution: Use correct format: 2024-FALL.
Example: Change `Fall 2024` to `2024-FALL`.

---

6 Grade Module Errors

6.1 Grade Not Found

Error: `"Grade not found."`
Cause: The specified grade ID doesn't exist in the system.
Solution: Verify the grade exists using Reports Transcript.
Example: Check ID 999 before updating.

6.2 Invalid Numeric Grade Range

Error: `"Invalid numeric grade. Must be between 0.0 and 100.0"`
Cause: Grade value is outside the 0-100 range.
Solution: Enter a grade between 0 and 100.
Example: Type `85.5`, not `105`.

6.3 Invalid Numeric Grade Format

Error: `"Invalid numeric grade. Must be 0-100."`
Cause: Grade value entered is invalid.

Solution: Enter a numeric value between 0 and 100.
Example: Type `85`, not `eighty-five`.

6.4 Enrollment Not Found for Grade

Error: `"Enrollment ID not found. (Internal Error)"`
Cause: Grade is being recorded for a non-existent enrollment.
Solution: Create the enrollment first, then record the grade.
Example: Enroll student first.

6.5 Grade Already Exists

Error: `"Grade already exists for this enrollment. Use 'Update Grade' instead."`
Cause: A grade has already been recorded for this enrollment.
Solution: Use Grade Management Update Grade to modify existing grade.
Example: Don't use 'Record New' for existing grades.

6.6 Student Not Found for Transcript

Error: `"Student not found."`
Cause: Student ID doesn't exist when generating transcript.
Solution: Verify the Student ID before viewing transcript.
Example: Search ID first.

6.7 Grade File Parse Error - Column Count

Error: `"[ERROR] Line X: Invalid column count. Skipped."`
Cause: Grade CSV row has wrong number of columns (expected 7).
Solution: Ensure each row matches the header structure.
Example: Correct: `1,1,2024001,1001,85.00,B+,2024-FALL`

6.8 Grade File Parse Error - Non-numeric Grade ID

Error: `"[ERROR] Line X: Grade ID 'A1' is not numeric. Skipped."`
Cause: Grade ID in CSV is not a number.
Solution: Use numeric IDs only.
Example: Change `A1` to `1`.

6.9 Grade File Parse Error - Invalid Numeric Value

Error: `"[ERROR] Line X: Numeric Grade '150.00' is not a valid number. Skipped."`
Cause: Numeric grade value in CSV is outside 0-100 range.
Solution: Correct the grade value to be within 0-100.
Example: Change `150.00` to `100.00`.

6.10 Grade File Parse Error - Duplicate Grade ID

Error: `"[ERROR] Line X: Duplicate Grade ID (1). Already loaded. Skipped."`
Cause: Same grade ID appears multiple times.
Solution: Each grade must have unique ID.
Example: Use unique IDs: 1, 2, 3...

6.11 Grade File Parse Error - Invalid Student ID

Error: `"[ERROR] Line X: Invalid Student ID (123). Skipped."`
Cause: Student ID in grade record doesn't match valid format.
Solution: Use 7-digit Student IDs in YYYYXXX format.
Example: Change `123` to `2024001`.

6.12 Grade File Parse Error - Invalid Grade Range

Error: `"[ERROR] Line X: Invalid Numeric Grade (105.00). Must be 0-100. Skipped."`
Cause: Numeric grade in CSV is outside valid range.
Solution: Correct grade to be between 0 and 100.
Example: Change `105.00` to `95.00`.

6.13 Grade File Parse Error - Enrollment Not Found

Error: `"[ERROR] Line X: Enrollment ID (999) not found in system. Cannot load grade. Skipped."`
Cause: Grade references an enrollment that doesn't exist.
Solution: Ensure enrollment exists before loading grade.
Example: Fix enrollment ID in grades.csv.

6.14 Grade File Parse Error - Invalid Semester

Error: `"[ERROR] Line X: Invalid Semester '2024 Fall'. Skipped."`
Cause: Semester format in CSV is incorrect.
Solution: Use YYYY-SEASON format.
Example: Change `2024 Fall` to `2024-FALL`.

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7 File Operation Errors

7.1 Students File Not Found

Error: `"Could not open students file."`
Cause: students.csv file is missing or inaccessible.
Solution: Create the file with correct header or check file permissions.
Example: Create empty `students.csv`.

7.2 Professors File Not Found

Error: `"Could not open professors file."`
Cause: professors.csv file is missing or inaccessible.
Solution: Create the file with correct header.
Example: Create empty `professors.csv`.

7.3 Courses File Not Found

Error: `"Could not open courses file."`
Cause: courses.csv file is missing or inaccessible.
Solution: Create the file with correct header.
Example: Create empty `courses.csv`.

7.4 Enrollments File Not Found

Error: `"Could not open enrollments file."`
Cause: enrollments.csv file is missing or inaccessible.
Solution: Create the file with correct header.
Example: Create empty `enrollments.csv`.

7.5 Grades File Not Found

Error: `"Could not open grades file."`
Cause: grades.csv file is missing or inaccessible.
Solution: Create the file with correct header.
Example: Create empty `grades.csv`.

7.6 File Save Error - Students

Error: `"Could not open file."`
Cause: Cannot write to students.csv file.
Solution: Close the file if open in another program, check write permissions.
Example: Close Excel if file is open.

7.7 File Save Error - Professors

Error: `"Could not open file for saving professors."`
Cause: Cannot write to professors.csv file.
Solution: Ensure file is not read-only and not open in another application.
Example: Uncheck "Read-only" in file properties.

7.8 File Save Error - Courses

Error: `"Could not open file for saving courses."`
Cause: Cannot write to courses.csv file.

Solution: Check file permissions and close other applications.
Example: Close text editors holding the file.

7.9 File Save Error - Enrollments

Error: `"Could not open file for saving enrollments."`
Cause: Cannot write to enrollments.csv file.
Solution: Verify disk space and file permissions.
Example: Check file locks.

7.10 File Save Error - Grades

Error: `"Could not open grades file for saving."`
Cause: Cannot write to grades.csv file.
Solution: Ensure the file path is valid and writable.
Example: Run as administrator if needed.

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8 Memory Errors

8.1 Initial Memory Allocation Failed

Error: `"Initial memory allocation failed."`
Cause: System cannot allocate initial memory for data arrays.
Solution: Close other applications to free memory, or restart the computer.
Example: Close web browsers freeing RAM.

8.2 Memory Reallocation Failed

Error: `"Memory reallocation failed."`
Cause: System cannot expand memory to accommodate more records.
Solution: Save current data immediately and restart the application.
Example: Save before data loss.

8.3 Memory Allocation Failed - Student

Error: `"Memory allocation failed!  Could not add student."`
Cause: Cannot allocate memory for new student record.
Solution: Save data, close other applications, and try again.
Example: Restart app then add student.

8.4 Memory Allocation Failed - Professor

Error: `"Memory allocation failed!  Could not add professor."`
Cause: Cannot allocate memory for new professor record.
Solution: Save, restart, retry.
Example: Free up RAM.

8.5 Memory Allocation Failed - Course

Error: `"Memory allocation failed."`
Cause: Cannot allocate memory for new course or prerequisite.
Solution: Free up system resources and try again.
Example: Close background apps.

8.6 Memory Allocation Failed - Enrollment

Error: `"Memory allocation failed."`
Cause: Cannot allocate memory for new enrollment.
Solution: Reduce memory usage and retry.
Example: Restart the system.

8.7 Memory Allocation Failed - Grade

Error: `"Memory allocation failed."`
Cause: Cannot allocate memory for new grade record.
Solution: Save current data and restart application.
Example: Restart program.

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9 CSV File Loading Errors

9.1 Invalid Column Count - General

Error: `"[ERROR] Line X: Invalid column count.  Skipped."`
Cause: CSV row doesn't have expected number of columns.
Solution: Check CSV file structure matches expected format.
Example: Count commas in the text editor.

9.2 CSV Parsing Checklist

When you see loading errors:

1. Open CSV in text editor (not Excel).
2. Check header row.
3. Count commas.
4. Look for quoted commas.
5. Check the line number mentioned in error.

10 Menu Navigation Errors

10.1 Invalid Menu Input

Error: `"Invalid input."`
Cause: Non-numeric input in menu selection.

Solution: Enter only the number corresponding to your choice.
Example: Type `1`, not `one`.

10.2 Invalid Menu Choice

Error: `"Invalid choice."`
Cause: Number entered doesn't correspond to any menu option.
Solution: Choose from the displayed menu options.
Example: If menu is 0-6, enter number in that range.

10.3 Test Results File Error

Error: `"[ERROR] Could not create test_results.txt"`
Cause: Test runner cannot create output file.
Solution: Ensure write permissions in project directory.
Example: Run as administrator.
